```
After mitigation metrics by group:
                    accuracy  false_positive_rate
sensitive_feature_0
GroupA              0.571429                 0.75
GroupB              0.500000                 1.00
```

Figure 2: Baseline result

```
metrics_mitigated = MetricFrame(
   metrics={"accuracy": accuracy_score, "false_positive_rate":
false_positive_rate},
   y_true=y_test,
   y_pred=y_pred_mitigated,
   sensitive_features=s_test
)

print("\nAfter mitigation metrics by group:")
print(metrics_mitigated.by_group)
```

Looking at the baseline results, we can see that the model behaves differently across the two groups. While accuracy is already uneven, the most striking issue is the false positive rate: both GroupA and GroupB have a false positive rate of 1.0, which means every negative example is incorrectly classified as positive. This is a clear signal that the baseline model is not just unfair, but also poorly calibrated for this dataset.

After applying demographic parity through in-processing mitigation, the results improve, at least for one group. GroupA sees a noticeable boost in accuracy (from 0.43 to 0.57) and a reduction in false positive rate (from 1.0 to 0.75). GroupB's accuracy stays the same, while its false positive rate remains unchanged at 1.0.

This highlights a key reality of fairness interventions: mitigation can reduce disparities, but it often comes with trade-offs and uneven gains across groups. Fairness constraints like demographic parity don't guarantee perfect balance or better performance everywhere, especially on small or noisy datasets. Instead, they provide a structured way to make bias visible and to reason explicitly about which kinds of fairness we care about, and which costs we're willing to accept.

In practice, this kind of analysis is less about finding a 'perfectly fair' model and more about making informed, transparent decisions about model behaviour before deployment.

This is just the beginning of your journey in responsible AI but can be a strong base for you to explore further and learn more in this field. END

By: Jishnu Saurav Mittapalli

The author is a full-stack web developer, interested in modern technologies like artificial intelligence and machine learning. He is currently working in the field of natural language processing.